

1.85inch ESP32-S3

TAIJI VIEWE PI Pro User Manual

Table of Contents

Product Overview	3
Version Control	4
1、 Purpose	4
2、 Core Version Difference Description	4
3、 Version Identification and Recognition Method	5
Product Functions	6
Screen Parameters	7
User Instructions	8
Network Configuration	8
Basic Operations	9
1、 Display Main Menu	9
2、 Operation of Each Function	9
Detailed User Instructions for Each Function	10
1、 Themed Clock	10
2、 Real-time Weather	10
3、 E-book	11
4、 Audio Spectrum	11
5、 Digital Photo Frame	11
6、 MJPEG Video	12
7、 MP3 Music	12
8、 G-force Meter	12
9、 Computer Secondary Screen (AIDA64)	13
10、 Settings	14
Product Display and Dimension Description	16
Product Image:	16
Dimension Drawing:	16

Product Overview

- ✧ Version Information: TAIJI VIEWE PI Pro-A (SmartRing-Plus-A) and TAIJI VIEWE PI Pro-B (SmartRing-Plus-B), see Version Control for details.
- ✧ Main Control Chip: ESP32-S3-N16R8 (dual-core MCU, integrated with Wi-Fi and Bluetooth BLE5.0, main frequency 240MHz, 520KB SRAM, 8MB PSRAM, 448KB ROM, 16MB Flash).
- ✧ Display Screen: 360×360 resolution, capacitive touch screen.
- ✧ Module Functions: Including LCD display, backlight control, touch screen control, I2S digital microphone, I2S digital-to-analog conversion, TF card interface, and speaker.
- ✧ Development Environment: Supports secondary development with Arduino IDE, ESP IDE, MicroPython, PlatformIO, etc., and UI development with LVGL.
- ✧ Manufacturing Process: CNC shell, exquisite and delicate in texture.

Version Control

1、 Purpose

This version control specification is formulated to standardize the version management of TAIJI VIEWE PI Pro (SmartRing-Plus) series products, clarify the core differences between products with different configurations, and ensure that the products can be accurately identified and managed in all links such as R&D, production, procurement, warehousing, and after-sales service.

2、 Core Version Difference Description

For the two models of TAIJI VIEWE PI Pro, all other hardware specifications (including but not limited to screen size, resolution, color gamut, whole machine structure, core functions, etc.) are completely consistent except for the different screen models and matching initialization parameters. The specific difference information is shown in the following table:

Product Version	Matching Screen Model	Wireless Charging	Speaker
TAIJI VIEWE PI Pro-A	VIEWE UE018HV-RB39-A002A	No	Yes
TAIJI VIEWE PI Pro-B	VIEWE UE018HV-RB39-A004A	No	Yes

The core difference between TAIJI VIEWE PI Pro models lies in the different screens (corresponding to different driver codes and

resolutions).

Note: TAIJI VIEWE PI Pro-A (SmartRing-Plus-A), TAIJI VIEWE PI Pro-B (SmartRing-Plus-B)

3、 Version Identification and Recognition Method

Product Nameplate Marking:

The product version model (SmartRing-Plus-A / SmartRing-Plus-B) is clearly marked on the product package for intuitive identification.

Firmware:

Before delivery, the product will be burned with firmware with version-matched initialization parameters, and the firmware version number suffix will distinguish the versions:

- (1) Firmware version suffix for SmartRing-Plus-A: _A
- (2) Firmware version suffix for SmartRing-Plus-B: _B

Software Update:

We will release sample codes of specific versions. Please select the corresponding reference according to the version for secondary development.

Product Functions

- Themed Clock: Automatically updates time after connecting to WIFI, with multiple built-in themed watch faces, and the time zone can be changed in Settings.
- Real-time Weather: Obtains weather information after connecting to the Internet via WIFI.
- E-book: Add text files to the SD card for novel reading and other purposes.
- Audio Spectrum: Multiple built-in styles for switching.
- Digital Photo Frame Function: Allows users to add their own photos.
- MJPEG Video: Allows users to add their own MJPEG files.
- MP3 Music Playback: Allows users to add their own MP3 files, supporting decoding of 320K high-quality MP3.
- Car G-force Watch Face: Built-in gyroscope to detect G-force based on the gyroscope sensor.
- Computer Secondary Screen (AIDA64): 5 built-in styles.
- Subsequent product function upgrades.

Screen Parameters

Name	Description	Unit	Remarks
Screen Type	IPS	—	—
Screen Size	1.85	Inch	Diagonal
Resolution	360*360	Dots	—
Display Area Size	45.68 (H) × 45.68 (V)	mm	—
Module Outline Size	48.08(H)×49.95(V)×2.12(T)	mm	
Driver IC	77916	—	
接口类型 Interface Type	QSPI	—	
TP IC	CST816S	—	—
TP Viewing	46.08 (H) × 46.08 (V)	mm	
TP Outline	55 (H) × 55 (V) × 0.7 (T)	mm	
Display Mode	Normal Black	—	
Brightness	400	cd/m ²	Typ.
Viewing Angle	ALL	0' clock	—
Display Color	262K	Colors	—
PPI	200	—	
Operating Temperature	-20 ~ 70	°C	
Storage Temperature	-30 ~ 80		
Operating Voltage	5	V	

User Instructions

Network Configuration

The device provides two network configuration methods: File-based Network Configuration and AP Network Configuration.

File-based Network Configuration:

Create a WIFI.txt file in the root directory of the memory card, write the WIFI name in the first line and the WIFI password in the second line of the file.

AP Network Configuration:

The AIDA64 secondary screen, real-time weather and real-time clock functions of this device require network configuration for use.

After power-on, the device will automatically turn on a hotspot (AP) named My-Ap with the password 12345678. Connect your mobile phone to this hotspot via WIFI, and a network configuration page will pop up after a short while. It will automatically search for WIFI in the current environment. Select the WIFI you want to connect to, enter the password, and click Connect to complete the network configuration. After successful configuration, the screen will obtain an IP address, which can be viewed in the Connection page under Settings.

Notes: The device can only connect to 2.4G band WIFI; some mobile phones will automatically disconnect from the hotspot and switch to 5G network once they detect that the hotspot cannot access the Internet, in which case you need to connect to the hotspot again. If the network configuration page does not pop up after connecting to the hotspot, you can also open it by entering 192.168.4.1 in the mobile phone browser.

Basic Operations

1、 Display Main Menu

The themed clock is the default interface after startup, and the main menu can be brought up by swiping up. The main menu can also be called up by swiping up in the weather, audio spectrum, digital photo frame, MJPEG playback, G-force watch face and AIDA64 interfaces. The main menu includes themed clock, weather, e-book, audio spectrum, digital photo frame, MJPEG playback, MP3, G-force watch face, AIDA64 secondary screen and Settings in the center; on the playlist screen of the MP3 and Settings pages, click the "○" icon at the top to call up the main menu; on the e-book page, click the "<" icon to call up the main menu; click the main menu to enter each function page.

2、 Operation of Each Function

- ✧ **Themed Clock:** Contains 4 themed watch face styles for clock display.
- ✧ **Real-time Weather:** Displays real-time weather conditions after connecting to WIFI, including location, time, date (including lunar calendar), temperature and humidity.
- ✧ **E-book:** Put text reading files into the txt folder of the memory card, and the device will read them automatically. Enter the e-book to read and turn pages by clicking the two sides of the screen.
- ✧ **Audio Spectrum:** Converts external sound sources into a beating spectrum, with 5 built-in audio spectrum themes.
- ✧ **Digital Photo Frame:** Swipe left or right on the screen to switch to the next or previous file for playback.
- ✧ **MJPEG Video:** Swipe left or right on the screen to switch to the next

or previous file for playback.

- ✧ **MP3:** Divided into playback page and menu/list page. The menu/list page is the default entry, which will automatically scan MP3 files in the music directory on the TF card and list them. Click the corresponding name to play, and switch to the playback page at the same time. Click any position on the screen on the playback page to display the menu/list page, which will be automatically hidden if there is no operation for 3 seconds.
- ✧ **Car G-force Watch Face:** Uses a gyroscope to sense balance and control the rolling direction of the ball. Single click to display the positioning icon, and long press to reposition the balance ball to the center of the screen.
- ✧ **Computer Secondary Screen (AIDA64):** Swipe left or right on the screen to switch between different secondary screen styles.
- ✧ **Settings:** All setting functions are performed on the Settings page. To extend the service life of the flash, modifications made on the Settings page will only be saved to the flash when exiting the Settings page.

Detailed User Instructions for Each Function

1、Themed Clock

Enter the clock interface. The Chinese version defaults to China Time Zone (UTC+8), and the international version defaults to International Time Zone (UTC+0). The time offset can be set in Settings > Language (LANGUAGE) > Timezone.

2、Real-time Weather

The function can be used after configuring the city code on the Web page.

Enter the city name on the web page and click Save to automatically generate the city code. However, it takes time to refresh. Exit the weather interface on the screen and then re-enter it. The corresponding city weather will be obtained after the network refresh is successful.

3、E-book

Store text files in the txt folder of the memory card, and the device will automatically identify and read them. After startup, click E-book to read, and turn pages by clicking the left and right arrows on the two edges of the screen. Night mode and page turning effects can be set through the Settings function.

4、Audio Spectrum

Gain calibration can be performed on the Web page with 4 levels in total. Adjust to an appropriate level to ensure that the spectrum is almost invisible in a quiet environment. There are 5 built-in different themes that can be switched by swiping left and right.

5、Digital Photo Frame

Create a folder named pic (lowercase, do not rename) in the root directory of the TF card. Put JPEG pictures into the pic folder. The picture resolution of 360*360 offers the best display effect without stretching and deformation. Insert the TF card into the device, and the device will automatically identify it after startup. Click the Digital Photo Frame function to display the pictures in the memory card. The display mode can be set through the Settings function, including sequential single image, sequential slide show, random single image and random slide show. The switching interval time can also be set.

6、 MJPEG Video

Convert the video to be played into an MJPEG file with the same screen resolution of 360*360, FPS of 25 and video quality of 7 through an MJPEG conversion tool, and copy the file to the mjpeg directory of the TF card. The playback repeat mode can be set through the Settings function: single loop, list loop and random play.

7、 MP3 Music

Put MP3 files with a bit rate below 320kbps and a sampling rate of 48000 or below into the music directory of the TF card (the device needs to be restarted after pulling out the card for copying and inserting it back). The playback repeat mode can be set through the Settings function: single loop, list loop and random play.

8、 G-force Meter

Using the G-force meter requires the resource file gmeter.jpg in the night7 directory of the TF card. If it is missing, please download it from the address we provide.

Record the device installation information for the first use. After placing the device and keeping it stationary, click the screen, then click the calibration button that appears in the center of the screen to calibrate, and keep the device stationary without shaking during the process.

It can be used normally after calibration.

For a better effect in daily driving, the G-force mapped by the middle ring is 0.3g.

9、Computer Secondary Screen (AIDA64)

1. Prepare an activated AIDA64 client (obtain it by yourself or contact us).
2. Open AIDA64, click [File] > [Settings] > [LCD] in the upper left corner, and select Remote Sensor. Note: General settings such as language can be made in the settings interface.
3. Set the port to 80, fill in a random resolution such as 1280×800, and check [Enable RemoteSensor LCD support].
4. Then click [LCD Items], click [Import] in the upper right corner, and select the aida_remote_1.85.rslcd file provided by us.
5. Manual settings are required after import, because everyone's CPU, GPU, motherboard, hard disk, network card, etc. are different, and settings need to be made item by item. The 8 parameters are CPU usage, CPU freq, CPU temp, CPU fan, GPU usage, GPU freq, GPU temp, GPU fan respectively. Click the 8 parameters at the bottom of the interface to set them correspondingly. **Note: Do not modify the text in "Show Label" under any circumstances, and "Show unit" must be filled with "" (without quotation marks), otherwise the device cannot detect it.**
6. Click OK after all modifications are completed, and then minimize AIDA64. You can set it to start automatically at boot (Settings > General > Run AIDA64 at Windows startup).
7. Access the IP address of the screen through a web page, set the IP address of your computer (i.e., the IP of the computer running AIDA64) in the "Secondary Screen Host Address" at the bottom, and then save it.

If the local port 80 is occupied, you need to add a colon and a custom

port after it.Example:

192.168.0.100: Use the host address of 192.168.0.100 with the default port 80.

192.168.0.100:9223: Use the host address of 192.168.0.100 with the custom port 9223.

Note: The colon is an English colon ":".

8. Allow AIDA64 to access the network in the firewall, or manually open the TCP 80 port (or your custom port). Important! Refer to the link for operation:

<https://jingyan.baidu.com/article/af9f5a2d2ea83543140a4584.html>

After the settings are completed, switch the screen to the AIDA64 function window to display the computer status information.

10、Settings

About:

It introduces the device manufacturer (VIEWE), the manufacturer's relevant information, device model, version and device memory status.

Language:

Provides two languages: English and Simplified Chinese. It also allows setting the time offset of the time zone and fonts. The currently optional fonts are Source Han Sans Bold, Source Han Sans, Source Han Serif Bold and Source Han Serif. It also supports custom fonts, which need to be imported into the fonts folder of the TF card.

System:

Set the screen brightness and screen rotation (clockwise rotation only in 90 ° increments); adjust the volume in audio settings; there are also

other settings, in which you can restart the device, clear settings, and enter engineering mode (engineering mode is only for testing the screen and touch, please restart the device to exit if needed).

Application:

Set the app to start automatically at boot; set the playback modes of music, pictures and MJPEG videos; set the night mode and page turning effects of e-books, etc.; the host IP address can be set in the AIDA secondary screen.

Connection:

Displays the IP address of the device, and provides settings for file-based network configuration, signal detection, restart to a specific mode, etc.

VIEWE TECHNOLOGY CO., LTD.

Product Display and Dimension Description

Product Image:



Dimension Drawing:

